

HEALTHCARE PATIENT VISIT ANALYZER - PROJECT REQUIREMENTS

PROJECT OBJECTIVES

“Analyze 100 patient visits to identify billing discrepancies, treatment patterns, and operational inefficiencies for St. Michael's Hospital revenue cycle management.”

DELIVERABLES (Complete These 4 Tasks)

TASK 1: DATA QUALITY AUDIT (CLEANING FOCUS)

1. Standardize all dates to MM/DD/YYYY format (4 different formats exist)
2. Fix diagnosis abbreviations (CAD→Coronary Artery Disease, etc.)
3. Clean "Notes" column - remove special characters, keep medical text only
4. Handle missing values in "CancellationReason" (mark as "Not Specified")
5. Ensure all currency amounts have 2 decimal places

TASK 2: FINANCIAL RECONCILIATION (FORMULA FOCUS)

1. Create "Calculated Patient Due" column:

text

= (VisitCost * (1 - InsuranceCoveragePct/100)) + CopayAmount + DeductibleApplied

2. Flag billing errors: If TotalPaid ≠ Calculated Patient Due, mark "CHECK"
3. Calculate insurance shortfall: VisitCost - (TotalPaid + InsuranceCoveragePct amount)

TASK 3: OPERATIONAL ANALYSIS (FILTERING FOCUS)

1. Filter to show only:
 - Medicare patients with visit cost > \$300
 - Rejected claims (BillingStatus = "Rejected")
 - No-show appointments (NoShowFlag = TRUE)
2. Create summary: Average satisfaction by department

TASK 4: CLINICAL REPORTING (LOOKUP FOCUS)

1. Create Doctor reference table (DoctorID, DoctorName, Specialty)
2. Create Diagnosis reference table (ICD-10 code, Full Diagnosis Name)

SUCCESS METRICS

1. Zero date format inconsistencies
2. All billing calculations accurate ($\pm \$0.01$)

DATA STRUCTURE (Your 100×40 dataset)

- Columns 1-10: Patient Demographics & Identification
- Columns 11-20: Visit Details & Clinical Data
- Columns 21-30: Financial Information
- Columns 31-40: Operational & Quality Metrics